



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

LSVCL302

SURVEYING COMMUNICATION LANGUAGE

Apply surveying communication language

Competence

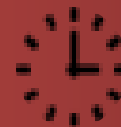
RQF Level: 3

Learning Hours

30

Credits: 3

Sector: Construction and Building Services



Trade: Land Surveying

Module Type: Specific/General/Complementary

Curriculum: CBSLSV3001- TVET Certificate 3 in Land Surveying

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Purpose statement	This module titled Apply surveying communication language was developed for students of RQF Level 3 for the purpose of equipping learners with skills, knowledge and right attitude through the application of surveying hand signals, symbols, abbreviations and surveying terms. Those signals and symbols will help the learners to effectively communicate in the field.					
Leaning assumed to be in place	Safety, Health and environment at workplace; Surveying tools and equipment.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	100%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual Work	50%				
			Summative Assessment		N.A	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria	
1. Apply surveying hand signals	1.1.	Types of hand signals are properly identified according to the information to be conveyed.
	1.2.	Surveying hand signals are properly described according to their types.
	1.3.	Surveying hand signals are adequately applied according to the types of information to be conveyed.
2. Apply Surveying symbols	2.1.	Types of surveying symbols are properly identified.
	2.2.	Surveying Symbols are properly described according to their types.

	2.3. Surveying Symbols are adequately applied according to the types of work.
3. Describe surveying terms and abbreviations	3.1. Basic Surveying terms are properly described according to the types of work.
	3.2. Basic Surveying abbreviations are properly identified.
	3.3. Use of surveying abbreviations is correctly described according to the types of surveying work.

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Apply Surveying hand signals 2. Apply Surveying symbols 3. Describe surveying terms and abbreviations
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Identify Surveying hand signals	Learning hours: 10
Indicative content ❖ Identification of types of hand signals • Hand signals for numerals • Hand and voice signals ❖ Description of Surveying hand signals • Hand signals for numerals • Hand and voice signals ❖ Application of surveying hand signals • Hand signals for Surveying measurements (Pegging, line ranging, building setting out, pipelines, power transmission line, drainage system, road setting out) • Hand signals for numerals (eg: Left arm extended diagonally up and shows a simple system for numerals out from, all right, move right or left, give the back sight, give a line or this is a hub, plumb the rod, establish a turning point) • Hand and voice signals	
Resources required for the learning outcome	

Equipment	Theodolite, levels, total station.
Materials	Papers, ink, lead mine, flip chart, notebook, pens, and pencils.
Tools	Flags, pegs, ranging rods, staff reading, ink pens, lead mine pens.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment • Practical assessment

Apply Surveying symbols	Learning hours: 10
Indicative content ❖ Identification of types of surveying symbols • Map symbols • Standard symbols • Topographic symbols • Convention symbols ❖ Description of surveying symbols • Map symbols • Standard symbols • Topographic map symbols (cultural symbols, natural features symbols, constructed features, hydrographical features, water features) • Convention symbol • Map legend • Surveyors symbols • Mineral survey symbols Application of surveying symbols	

• Marsh or swamp, house, embankment, single line railway, lake or pond, road, railway bridge, river and fence. • Demonstration of symbols on maps and plans • Locate and differentiate symbols on surveying field	
Resources required for the indicative content	
Equipment	GPS receiver, DGNS receiver, level, ODM, DDM, EDM.
Materials	Papers spray paint, nails, pegs.
Tools	Plans, maps, pencils, pens, marker pens, pins, flags, monuments, beacons, claw hammers, machete, pick axe, plumb bob, tape measures.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment

Describe basic surveying terms and abbreviations	Learning hours: 10
Indicative content ❖ Description of basic surveying terms • Topographic surveying terms (mapping, contour lines, traversing, etc.) • Cadastral surveying terms (land subdivision, registration, cadastre, etc.) • Levelling terms (Mean sea level, datum, benchmark, reduced level, etc.) • Angular measurements terms (bearing, zenith, azimuth, etc.) • Linear measurements terms (direct and indirect, slope, height, etc.) ❖ Identification of basic surveying abbreviations • Linear measurements abbreviations (EDM, PK, etc.) • Levelling surveying abbreviations (BM, RL, CP, etc.)	

• Angular measurements abbreviations (WCB, RB, etc.) • Topographic surveying abbreviations (TBM, PK, HI, etc.) ❖ Application domain of basic surveying abbreviations • Road construction surveying • Topographic surveying • Angular measurements • Levelling surveying • Linear measurements • Cadastral surveying • Cartography • Construction surveying • Mining surveying	
Resources required for the indicative content	
Equipment	GPS, DGPS, total station, level, theodolite, EDM.
Materials	Papers, spray paint, nails, pegs.
Tools	Plans, maps, pencils, pens, marker pens, pins, flags, monuments, beacons, claw hammers, machete, pick axe, plumb bob, tape measures.
Facilitation techniques	• Individual and group work • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment

References:

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